

From Mechanically Interlocked Structures to Host–Guest Chemistry Based on Twisted Dimeric Architectures by Adjusting Space Constraints

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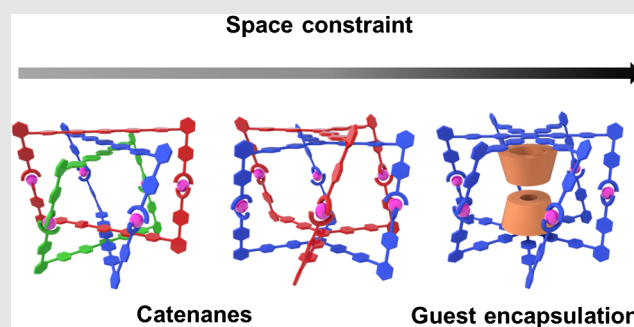
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Mechanically interlocked molecules (MIMs) and host–guest chemistry have received great attention in the past few decades. However, it remains challenging to design architectures with mechanically interlocked features and construct cavities for guest molecule recognition using similar building blocks. In this study, we designed and constructed a series of novel twisted supramolecular structures by assembling various multitopic terpyridine (tpy) ligands with the same diameter and Zn(II) ions. The obtained complexes exhibited evolutionary architectures and showed distinctively different space-constraint effects. Specifically, the assembled dimer SA, SB, and SBH displayed mechanically interlocked phenomena, including [2]catenane and [3]catenane, with an increase in concentration. However, no interlocked structures were observed in complexes SC and SCH constructed by hexatopic tpy ligands due to the significant space

constraints. The single-crystal data of complex SCH further proved significant space constraints and illustrated the formation of a relatively closed cavity, which showed excellent host–guest properties for different calixarenes, especially high affinity for calix[6]arene.



Keywords: self-assembly, terpyridine, mechanical interlocking, host–guest, space constraints

Introduction

Inspired by the self-assembly phenomena and functional expressions of natural molecules, such as DNA formation, protein folding or recognition, and enzyme catalysis,^{1–6} noncovalent molecular interactions, including metal-ligand coordination, hydrogen bonding, π - π interactions, and so on,^{7–9} have been extensively applied to construct numerous discrete architectures.^{10–12} Based on the discrete structures, more complex systems formed by intermolecular interactions, such as hierarchical self-assembly,^{13–15} the mechanically interlocked molecules (MIMs),^{16–19} and host-guest systems,^{20–24} have received considerable attention over the past few decades. As widespread structures playing a unique role in living organisms, complex systems inspire the fabrication of aesthetic molecular structures and provide impetus to the development of molecular machines and smart materials.^{25–30}

Coordination-driven self-assembly has served as one of the most effective approaches to construct complex and exquisite discrete architectures, due to the feature of dynamic reversibility and coordination geometry.^{31–37} In recent years, by employing discrete metallosupramolecular structures, a variety of interlocked architectures have been extensively reported, including rotaxanes,^{38–40} catenanes,^{41–44} and knots.^{45–48} In contrast, many discrete metallosupramolecules have been widely employed in the fabrication of functional host-guest systems, such as catalysis,^{49,50} molecular binding,⁵¹ and drug delivery.^{52,53} For interlocked and host-guest molecular systems, an appropriate cavity with predesigned interactions and space constraints is the most critical factor for fabricating the interlocked structures and guest encapsulation. Although many types of cavities have been created for complex molecular systems, developing a suitable system to transform interlocked to host-guest structures by simply adjusting space constraints remains a great challenge.

2,2':6',2''-Terpyridine derivatives have been widely used to construct numerous novel supramolecular structures^{54–59} with special applications in diverse fields, including catalysts,^{60,61} medicines,⁶² sensors,^{63,64} smart materials,^{65,66} and optoelectronic devices.^{67–69} However, the octahedral coordination geometry of the tpy-M(II)-tpy complex can cause significant steric hindrance, which limits the application in the formation of coordination-directed, instead of templating-directed, mechanically interlocked structures.^{70–73} On the contrary, the high steric hindrance of tpy-M(II)-tpy is more advantageous to construct a tighter closed cavity for the encapsulation of guest molecules.

Herein, we designed and synthesized five ortho- tpy ligands with the same diameter (Scheme 1), denoted as ditopic **LA**, tritopic **LB** and **LBH**, and hexatopic **LC** and **LCH**. To reduce the steric congestion of the

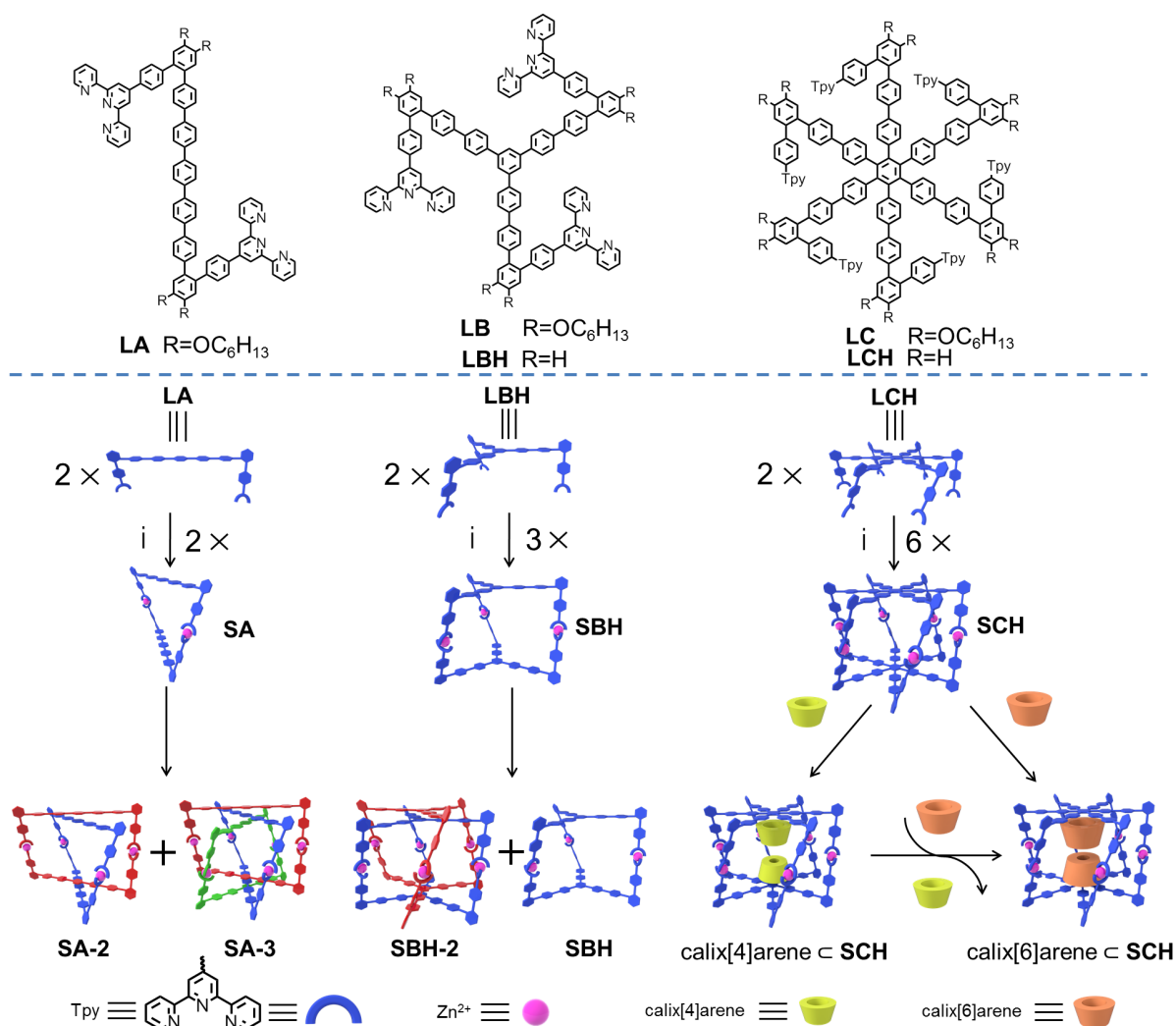
pseudo-octahedral tpy-M(II)-tpy connection, tpy units were located in the center of the final structures. The corresponding complexes with twisted dimeric structures were successfully constructed through assembling these ligands with Zn(II) ions. The obtained complexes exhibited different space constraints and showed significantly different interlocking or host-guest properties. For metal complexes **SA**, **SB**, and **SBH**, assembled by ditopic ligand **LA**, and tritopic ligands **LB** and **LBH** (without alkyl chains), the obvious self-interlocked phenomena were observed upon increasing the concentration of these complexes, due to their lower space constraints and suitable cavity. However, with further increasing space constraints, the complexes **SC** and **SCH**, based on the hexatopic ligands, could only form monomeric twisted prisms, and no interlocking phenomena were observed. Because of the significant space restriction effect, **SCH** served as a well-sealed cavity for the host-guest properties and efficiently encapsulated the calixarene-type species, especially showing a high affinity for calix[6]arene, with a more complex conformation.

Experimental Methods

All reagents were purchased from Sigma-Aldrich (Chaoyang District, Beijing, China), Acros (Shanghai, China), and Aladdin (Shanghai, China) and used without further purification. Compounds **1**,⁷⁴ **2**,⁷⁵ **4**,⁷⁶ **5**,⁷³ **7**,⁷⁷ and **8**⁷⁸ were synthesized per the literature as reported in Supporting Information Scheme S1. NMR data were recorded at 25 °C on Bruker 500 MHz and 600 MHz (Bruker Switzerland AG, Fällanden, Zurich, Switzerland) nuclear magnetic resonance instruments using CDCl_3 and CD_3CN with tetramethylsilane (TMS) as the solvents. Electrospray ionization mass spectrometry (ESI-MS) and traveling-wave ion mobility mass spectrometry (TWIM-MS) were recorded with a Waters Synapt G2 tandem mass spectrometer (Waters Corporation, Milford, MA, USA). Matrix-assisted laser desorption ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was performed on a Bruker Autoflex III (Bruker Corporation, Billerica, MA, USA) using trans-2-[3-(4-tert-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB) as a matrix. Crystal data were collected from the BL17B beamline of National Facility for Protein Science in Shanghai (NFPS) at Shanghai Synchrotron Radiation Facility.

Synthesis of ligand LA

Compound **6** (0.9 g, 1.14 mmol), 4,4''-dibromo-p-terphenyl (186.4 mg, 0.48 mmol), $\text{Pd(PPh}_3)_2\text{Cl}_2$ (40 mg, 0.058 mmol), and sodium carbonate (1.59 g, 15.0 mmol) were added into a 100 ml Schlenk flask. Toluene (30 mL), H_2O (15 mL), and *tert*-butyl alcohol (6 mL) were added under N_2 . The mixture was stirred at 85 °C for 24 h. After cooling to room temperature, the solvent was removed under vacuum, and the residue was extracted with CHCl_3 .



Scheme 1 | The self-assembly of twisted dimeric architectures, mechanically interlocked structures, and guest recognition. Conditions: (1) $CHCl_3/MeOH$ (1:3, v/v), 50 °C, 8 h.

The combined organic layer was washed with brine, dried over anhydrous Na_2SO_4 , and then concentrated in vacuo. The crude product was purified by column chromatography on silica gel with dichloromethane: ethanol (100:2) as eluent to afford the product as a white solid (0.49 g, 65%). 1H NMR (500 MHz, $CDCl_3$, 298 K, δ): 8.73 (s, 4H, tpy- $H^{3',5'}$), 8.72–8.69 (m, 4H, tpy- $H^{6,6'}$), 8.66 (d, $J = 7.9$ Hz, 4H, tpy- $H^{3,3'}$), 7.87 (td, $J = 7.7, 1.8$ Hz, 4H, tpy- $H^{4,4'}$), 7.80 (d, $J = 8.2$ Hz, 4H, Ph- H^i), 7.70 (d, $J = 4.6$ Hz, 12H, Ph- H^m , Ph- H^o , and Ph- H^n), 7.55 (d, $J = 8.0$ Hz, 4H, Ph- H^j), 7.36–7.31 (m, 8H, tpy- $H^{5,5'}$ and Ph- H^l), 7.27 (m, 4H, Ph- H^k), 7.03 (d, $J = 2.2$ Hz, 4H, Ph- H^a and Ph- H^b), 4.11 (td, $J = 6.6, 3.9$ Hz, 8H, Alkyl- H^c and Alkyl- $H^{c'}$), 1.93–1.88 (m, 8H, Alkyl- H^d and Alkyl- $H^{d'}$), 1.52 (8H, Alkyl- H^e and Alkyl- $H^{e'}$), 1.38 (m, 16H, Alkyl- H^f , Alkyl- H^g , Alkyl- H^f , and Alkyl- H^g), 0.96–0.89 (m, 12H, Alkyl- H^h and Alkyl- $H^{h'}$). ^{13}C NMR (125 MHz, $CDCl_3$, 298 K, δ): 156.40, 156.00, 150.04, 149.24, 148.78, 148.65, 142.60, 140.64, 139.77, 139.51, 138.53, 137.01, 136.25, 132.86,

132.51, 130.66, 130.54, 127.55, 127.50, 127.42, 127.06, 126.72, 123.95, 121.48, 118.83, 116.20, 31.77, 29.45, 25.90, 22.81, 14.23, 0.16. MALDI-TOF-MS (m/z): calcd for $[C_{108}H_{104}N_6O_4]^+$, 1548.8; found, 1548.8.

Synthesis of ligand LB

Compound **6** (1.0 g, 1.27 mmol), 1,3,5-tris(4-bromophenyl) benzene (178 mg, 0.33 mmol), $Pd(PPh_3)_2Cl_2$ (44 mg, 0.064 mmol), and sodium carbonate (1.06 g, 10.0 mmol) were added into a 100 ml Schlenk flask. Toluene (25 mL), H_2O (10 ml), and *tert*-butyl alcohol (3 mL) were added under N_2 . The mixture was stirred at 85 °C for 3 days. After cooling to room temperature, the solvent was removed under vacuum, and the residue was extracted with $CHCl_3$. The combined organic layer was washed with brine, dried over anhydrous Na_2SO_4 , and then concentrated in vacuo. The crude product was purified by column

chromatography on silica gel with dichloromethane: ethanol (100:2.5) as eluent to afford the product as a white solid (0.51 g, 68%). ¹H NMR (500 MHz, CDCl₃, 298 K, δ): 8.73 (s, 6H, tpy-*H*^{3,5}), 8.71–8.68 (m, 6H, tpy-*H*^{6,6'}), 8.65 (dd, *J* = 7.9, 1.1 Hz, 6H, tpy-*H*^{3,3'}), 7.86–7.83 (m, 9H, tpy-*H*^{4,4'}, Ph-*H*^o), 7.82–7.78 (m, 6H, Ph-*H*ⁱ), 7.77–7.71 (m, 12H, Ph-*H*ⁿ and Ph-*H*^m), 7.58–7.52 (m, 6H, Ph-*H*^j), 7.35–7.30 (m, 12H, Ph-*H*^k and tpy-*H*^{5,5'}), 7.27 (m, 6H, Ph-*H*^k), 7.03 (d, *J* = 1.4 Hz, 6H, Ph-*H*^p and Ph-*H*^q), 4.12 (td, *J* = 6.6, 4.4 Hz, 12H, Alkyl-*H*^c and Alkyl-*H*^{c'}), 1.88 (m, 12H, Alkyl-*H*^d and Alkyl-*H*^{d'}), 1.52 (m, 12H, Alkyl-*H*^e and Alkyl-*H*^{e'}), 1.38 (m, 24H, Alkyl-*H*^f, Alkyl-*H*^g, Alkyl-*H*^{f'}, and Alkyl-*H*^{g'}), 0.95–0.89 (m, 18H, Alkyl-*H*^h and Alkyl-*H*^{h'}). ¹³C NMR (125 MHz, CDCl₃, 298 K, δ): 156.35, 155.98, 150.01, 149.22, 148.76, 148.63, 142.58, 142.08, 140.67, 139.99, 138.50, 137.00, 136.23, 132.83, 132.48, 130.65, 130.54, 127.78, 127.58, 127.05, 126.75, 125.03, 123.95, 121.46, 118.81, 116.18, 31.76, 29.44, 25.89, 22.80, 14.22. MALDI-TOF-MS (*m/z*): calcd for [C₁₅₉H₁₅₃N₉O₆+H]⁺, 2285.2; found, 2285.2.

Synthesis of ligand LBH

Compound **3** (1.2 g, 2.0 mmol), 1,3,5-tris(4-bromophenyl) benzene (0.31 g, 0.57 mmol), Pd(PPh₃)₂Cl₂ (70 mg, 0.1 mmol), and sodium carbonate (1.59 g, 15 mmol) were added into a 100 mL Schlenk flask. Toluene (30 mL), H₂O (15 mL), and *tert*-butyl alcohol (5 mL) were added under N₂. The mixture was stirred at 85 °C for 3 days. After cooling to room temperature, the solvent was removed under vacuum, and the residue was extracted with CHCl₃. The combined organic layer was washed with brine, dried over anhydrous Na₂SO₄, and then concentrated in vacuo. The crude product was purified by column chromatography on silica gel with dichloromethane: ethanol (100:2.5) as eluent to afford the product as a white solid (0.62 g, 65%). ¹H NMR (500 MHz, CDCl₃, 298 K, δ): 8.73 (s, 6H, tpy-*H*^{3,5}), 8.69 (dd, *J* = 4.9, 1.7 Hz, 6H, tpy-*H*^{6,6'}), 8.65 (d, *J* = 7.9 Hz, 6H tpy-*H*^{3,3'}), 7.87–7.79 (m, 15H, tpy-*H*^{4,4'}, Ph-*H*^k and Ph-*H*^p), 7.75 (m, 12H, Ph-*H*ⁱ and Ph-*H*^j), 7.57 (d, *J* = 8.0 Hz, 6H, Ph-*H*^h), 7.54–7.52 (m, 6H, Ph-*H*^d and Ph-*H*^c), 7.51–7.46 (m, 6H, Ph-*H*^f and Ph-*H*^e), 7.35 (d, *J* = 8.2 Hz, 6H, Ph-*H*^b), 7.30 (m, 12H, tpy-*H*^{5,5'} and Ph-*H*^q). ¹³C NMR (125 MHz, CDCl₃, δ): 156.38, 156.02, 149.99, 149.23, 142.55, 142.09, 140.61, 140.31, 140.05, 138.85, 136.98, 136.61, 130.86, 130.59, 130.49, 127.97, 127.79, 127.60, 127.10, 126.78, 125.04, 123.94, 121.47, 118.87, 77.42, 77.16, 76.91. MALDI-TOF-MS (*m/z*): calcd for [C₁₂₃H₈₁N₉]⁺, 1683.7; found, 1683.7.

Synthesis of ligand LC

Compound **6** (1.0 g, 1.27 mmol), hexa(4-bromophenyl)-benzene (163 mg, 0.16 mmol), Pd(PPh₃)₄ (72 mg, 0.064 mmol), and sodium carbonate (1.59 g, 15 mmol) were added into a 100 mL Schlenk flask. Toluene (30 mL), H₂O (15 mL), and *tert*-butyl alcohol (5 mL) were added

under N₂. The mixture was stirred at 85 °C for 6 days. After cooling to room temperature, the solvent was removed under vacuum, and the residue was extracted with CHCl₃. The combined organic layer was washed with brine, dried over anhydrous Na₂SO₄, and then concentrated in vacuo. The crude product was purified by column chromatography on silica gel with dichloromethane: ethanol (100:3.5) as eluent to afford the product as a white solid (251 mg, 35%). ¹H NMR (500 MHz, CDCl₃, 298 K, δ): 8.60 (s, 12H, tpy-*H*^{3,5}), 8.56 (d, *J* = 4.8 Hz, 12H, tpy-*H*^{6,6'}), 8.52 (d, *J* = 8.0 Hz, 12H, tpy-*H*^{3,3'}), 7.74 (td, *J* = 7.8, 1.8 Hz, 12H, tpy-*H*^{4,4'}), 7.63 (d, *J* = 7.9 Hz, 12H, Ph-*H*ⁱ), 7.28 (m, 12H, Ph-*H*ⁿ), 7.16 (m, 36H, tpy-*H*^{5,5'}, Ph-*H*^j and Ph-*H*^k), 7.00 (d, *J* = 8.2 Hz, 12H, Ph-*H*^m), 6.93 (s, 6H, Ph-*H*^p), 6.89–6.83 (m, 18H, Ph-*H*^k and Ph-*H*^b), 4.08–3.96 (m, 24H, Alkyl-*H*^c and Alkyl-*H*^{c'}), 1.87–1.77 (m, 24H, Alkyl-*H*^d and Alkyl-*H*^{d'}), 1.51–1.42 (m, 24H, Alkyl-*H*^e and Alkyl-*H*^{e'}), 1.39–1.29 (m, 48H, Alkyl-*H*^f, Alkyl-*H*^g, Alkyl-*H*^{f'}, and Alkyl-*H*^{g'}), 0.92–0.83 (m, 36H, Alkyl-*H*^h and Alkyl-*H*^{h'}). ¹³C NMR (125 MHz, CDCl₃, δ): 156.16, 155.70, 149.55, 148.95, 148.61, 148.35, 142.32, 139.88, 138.53, 136.68, 135.78, 132.81, 131.83, 130.35, 130.00, 129.45, 126.74, 126.34, 125.23, 125.22, 123.64, 121.21, 118.50, 116.25, 116.24, 115.63, 31.59, 29.28, 25.73, 22.61, 14.03. MALDI-TOF-MS (*m/z*): calcd for [C₃₁₂H₃₀₀N₁₈O₁₂+H]⁺, 4491.3; found, 4491.3.

Synthesis of ligand LCH

Compound **3** (1.5 g, 2.6 mmol), hexa(4-bromophenyl)-benzene (328 mg, 0.33 mmol), Pd(PPh₃)₄ (150 mg, 0.13 mmol), and sodium carbonate (1.59 g, 15 mmol) were added into a 100 mL Schlenk flask. Toluene (30 mL), H₂O (15 mL), and *tert*-butyl alcohol (4 mL) were added under N₂. The mixture was stirred at 85 °C for 6 days. After cooling to room temperature, the solvent was removed under vacuum, and the residue was extracted with CHCl₃. The combined organic layer was washed with brine, dried over anhydrous Na₂SO₄, and then concentrated in vacuo. The crude product was purified by column chromatography on silica gel with dichloromethane: ethanol (100:3.5) as eluent to afford the product as a white solid (0.47 g, 43%). ¹H NMR (500 MHz, CDCl₃, 298 K, δ): 8.59 (s, 12H, tpy-*H*^{3,5}), 8.58–8.55 (m, 12H, tpy-*H*^{6,6'}), 8.51 (d, *J* = 8.0 Hz, 12H, tpy-*H*^{3,3'}), 7.74 (td, *J* = 7.7, 1.8 Hz, 12H, tpy-*H*^{4,4'}), 7.64 (d, *J* = 8.1 Hz, 12H, Ph-*H*^p), 7.43–7.34 (m, 18H, Ph-*H*^c, Ph-*H*^d, and Ph-*H*^e), 7.30 (d, *J* = 7.4 Hz, 6H, Ph-*H*^f), 7.29 (s, 12H, Ph-*H*^j), 7.19 (ddd, *J* = 7.5, 4.7, 1.2 Hz, 12H, tpy-*H*^{5,5'}), 7.15 (d, *J* = 8.0 Hz, 12H, Ph-*H*^b), 7.11 (d, *J* = 8.0 Hz, 12H, Ph-*H*ⁱ), 6.99 (d, *J* = 8.1 Hz, 12H, Ph-*H*^j), 6.86 (d, *J* = 8.0 Hz, 12H, Ph-*H*^q). ¹³C NMR (125 MHz, CDCl₃, δ): 156.26, 155.82, 149.66, 149.09, 142.44, 140.36, 140.25, 139.90, 139.83, 138.97, 137.07, 136.84, 136.24, 131.96, 130.94, 130.78, 130.41, 130.08, 127.82, 127.10, 126.92, 126.50, 125.41, 123.80, 121.38, 118.68. MALDI-TOF-MS (*m/z*): calcd. for [C₂₄₀H₁₅₆N₁₈]⁺, 3289.3; found, 3289.3

Synthesis of complex SA

To a solution of ligand **LA** (10.0 mg, 6.5 μmol) in CHCl_3 (3 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.9 mg, 6.4 μmol) in MeOH (9 mL) was added. The mixture was stirred at 50 °C for 8 h and then cooled to room temperature. Upon addition of NH_4PF_6 (100 mg), a precipitate was formed and washed with water to give a white product (10.6 mg). ^1H NMR (500 MHz, CD_3CN , 298 K, δ): 8.95 (s, 8H, $\text{tpy-H}^{3,5}$), 8.68 (d, $J = 8.1$ Hz, 8H, $\text{tpy-H}^{3,3'}$), 8.10–8.14 (m, 16H, $\text{tpy-H}^{4,4'}$, Ph- H^j), 7.79 (m, 32H, $\text{tpy-H}^{6,6'}$, Ph- H^m , Ph- H^n , and Ph- H^o), 7.65 (d, $J = 8.0$ Hz, 8H, Ph- H^i), 7.57 (d, $J = 8.1$ Hz, 8H, Ph- H^j), 7.36 (m, 16H, Ph- H^k and $\text{tpy-H}^{5,5'}$), 7.13 (s, 4H, Ph- H^p), 7.10 (s, 4H, Ph- H^q), 4.14 (q, $J = 6.9$ Hz, 16H, Alkyl- H^c and Alkyl- $\text{H}^{c'}$), 1.84 (m, 16H, Alkyl- H^d and Alkyl- $\text{H}^{d'}$), 1.53 (m, 16H, Alkyl- H^e and Alkyl- $\text{H}^{e'}$), 1.42–1.36 (m, 32H, Alkyl- H^f , Alkyl- H^g , Alkyl- H^h , and Alkyl- $\text{H}^{h'}$). ^{13}C NMR (125 MHz, CD_3CN , 298 K, δ): 155.50, 149.77, 149.05, 148.62, 147.94, 147.84, 144.94, 141.63, 141.14, 140.79, 139.64, 139.57, 138.16, 133.55, 132.72, 131.60, 131.25, 130.69, 127.84, 127.48, 127.43, 127.22, 126.43, 124.66, 123.16, 120.99, 116.11, 31.34, 29.13, 25.53, 22.40, 13.36. ESI-MS (m/z): 1760.3 [$\text{M}-2\text{PF}_6^-$] $^{2+}$ (calcd: 1760.3), 1125.0 [$\text{M}-3\text{PF}_6^-$] $^{3+}$ (calcd: 1125.0), 807.5 [$\text{M}-4\text{PF}_6^-$] $^{4+}$ (calcd: 807.5).

Synthesis of complex SB

To a solution of ligand **LB** (8.0 mg, 3.5 μmol) in CHCl_3 (3 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.6 mg, 5.4 μmol) in MeOH (9 mL) was added. The mixture was stirred at 50 °C for 8 h and then cooled to room temperature. Upon addition of NH_4PF_6 (100 mg), a precipitate was formed and washed with water to give a white product (9.1 mg). ^1H NMR (500 MHz, CD_3CN , 298 K, δ): 8.94 (s, 12H, $\text{tpy-H}^{3,5}$), 8.67 (d, $J = 8.1$ Hz, 12H, $\text{tpy-H}^{3,3'}$), 8.11 (d, $J = 7.8$ Hz, 24H, $\text{tpy-H}^{4,4'}$ and Ph- H^j), 7.98–7.86 (m, 18H, Ph- H^o and Ph- H^n), 7.78 (m, 24H, Ph- H^m and Ph- H^i), 7.63 (d, $J = 7.7$ Hz, 12H, $\text{tpy-H}^{6,6'}$), 7.56 (d, $J = 8.0$ Hz, 12H, Ph- H^i), 7.34 (m, 24H, Ph- H^k and $\text{tpy-H}^{5,5'}$), 7.14 (s, 6H, Ph- H^p), 7.09 (s, 6H, Ph- H^q), 4.19–4.06 (m, 24H, Alkyl- H^c and Alkyl- $\text{H}^{c'}$), 1.84 (s, 24H, Alkyl- H^d and Alkyl- $\text{H}^{d'}$), 1.53 (s, 24H, Alkyl- H^e and Alkyl- $\text{H}^{e'}$), 1.42–1.36 (m, 48H, Alkyl- H^f , Alkyl- H^g , Alkyl- H^h and Alkyl- $\text{H}^{h'}$). ESI-MS (m/z): 1734.1 [$\text{M}-3\text{PF}_6^-$] $^{3+}$ (calcd: 1734.1), 1264.5 [$\text{M}-4\text{PF}_6^-$] $^{4+}$ (calcd: 1264.5), 982.4 [$\text{M}-5\text{PF}_6^-$] $^{5+}$ (calcd: 982.4), 794.5 [$\text{M}-6\text{PF}_6^-$] $^{6+}$ (calcd: 794.5).

Synthesis of complex SBH

To a solution of ligand **LBH** (10.0 mg, 5.9 μmol) in CHCl_3 (3 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (2.6 mg, 8.9 μmol) in MeOH (9 mL) was added. The mixture was stirred at 50 °C for 8 h and then cooled to room temperature. Upon addition of NH_4PF_6 (100 mg), a precipitate was formed and washed with water to give a white product (11.8 mg). ^1H NMR (500 MHz, CD_3CN , 298 K, δ): 8.95

(s, 12H, $\text{tpy-H}^{3,5}$), 8.67 (d, $J = 8.1$ Hz, 12H, $\text{tpy-H}^{3,3'}$), 8.15–8.06 (m, 24H, Ph- H^p and $\text{tpy-H}^{4,4'}$), 7.96 (s, 6H, Ph- H^k), 7.90 (d, $J = 8.0$ Hz, 12H, Ph- H^j), 7.78 (m, 24H, $\text{tpy-H}^{6,6'}$ and Ph- H^i), 7.66 (d, $J = 8.0$ Hz, 12H, Ph- H^n), 7.63–7.56 (m, 36H, Ph- H^d , Ph- H^e , Ph- H^f , Ph- H^b , and Ph- H^f), 7.38 (d, $J = 8.0$ Hz, 12H, Ph- H^p), 7.34 (dd, $J = 7.7, 5.2$ Hz, 12H, $\text{tpy-H}^{5,5'}$). ^{13}C NMR (125 MHz, CD_3CN , 298 K, δ): 149.79, 147.96, 144.92, 141.62, 141.16, 140.77, 140.04, 139.24, 138.49, 136.69, 135.19, 134.83, 134.01, 133.67, 131.14, 130.70, 130.58, 128.53, 128.07, 127.84, 127.58, 127.44, 127.26, 126.49, 124.67, 123.17, 121.11. ESI-MS (m/z): 1333.6 [$\text{M}-3\text{PF}_6^-$] $^{3+}$ (calcd: 1333.6), 963.7 [$\text{M}-4\text{PF}_6^-$] $^{4+}$ (calcd: 963.7), 742.0 [$\text{M}-5\text{PF}_6^-$] $^{5+}$ (calcd: 742.0), 594.1 [$\text{M}-6\text{PF}_6^-$] $^{6+}$ (calcd: 594.1).

Synthesis of complex SC

To a solution of ligand **LC** (6.0 mg, 1.34 μmol) in CHCl_3 (2 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.2 mg, 4.0 μmol) in MeOH (6 mL) was added. The mixture was stirred at 50 °C for 8 h and then cooled to room temperature. Upon addition of NH_4PF_6 (80 mg), a precipitate was formed and washed with water to give a white product (6.8 mg, 92%). ^1H NMR (500 MHz, CD_3CN , 298 K, δ): 8.74 (s, 24H, $\text{tpy-H}^{3,5}$), 8.59 (d, $J = 8.0$ Hz, 24H, $\text{tpy-H}^{3,3'}$), 8.04 (t, $J = 7.6$ Hz, 24H, $\text{tpy-H}^{4,4'}$), 7.70 (d, $J = 5.0$ Hz, 24H, $\text{tpy-H}^{6,6'}$), 7.61 (d, $J = 7.8$ Hz, 24H, Ph- H^j), 7.44–7.39 (m, 48H, Ph- H^n and Ph- H^i), 7.30–7.21 (m, 72H, Ph- H^k , Ph- H^l , and $\text{tpy-H}^{5,5'}$), 7.15 (d, $J = 7.9$ Hz, 24H, Ph- H^m), 6.97 (m, 24H, Ph- H^p and Ph- H^q), 4.05 (s, 48H, Alkyl- H^c and Alkyl- $\text{H}^{c'}$), 1.77 (d, $J = 8.7$ Hz, 48H, Alkyl- H^d and Alkyl- $\text{H}^{d'}$), 1.46 (s, 48H, Alkyl- H^e and Alkyl- $\text{H}^{e'}$), 1.33 (s, 96H, Alkyl- H^f , Alkyl- H^g , Alkyl- H^h , and Alkyl- $\text{H}^{h'}$). ^{13}C NMR (125 MHz, CD_3CN , 298 K, δ): 156.28, 155.59, 154.36, 148.69, 147.74, 140.82, 140.33, 140.28, 137.96, 135.66, 135.08, 132.48, 130.42, 129.12, 127.34, 127.22, 126.96, 126.00, 125.76, 122.80, 31.33, 29.16, 25.53, 25.30, 13.16. ESI-MS (m/z): 2079.1 [$\text{M}-5\text{PF}_6^-$] $^{5+}$ (calcd: 2079.1), 1708.2 [$\text{M}-6\text{PF}_6^-$] $^{6+}$ (calcd: 1708.2), 1443.6 [$\text{M}-7\text{PF}_6^-$] $^{7+}$ (calcd: 1443.6), 1244.9 [$\text{M}-8\text{PF}_6^-$] $^{8+}$ (calcd: 1244.9), 1090.6 [$\text{M}-9\text{PF}_6^-$] $^{9+}$ (calcd: 1090.6), 966.8 [$\text{M}-10\text{PF}_6^-$] $^{10+}$ (calcd: 966.8).

Synthesis of complex SCH

To a solution of ligand **LCH** (6.0 mg, 1.82 μmol) in CHCl_3 (2 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.6 mg, 5.4 μmol) in MeOH (6 mL) was added. The mixture was stirred at 50 °C for 8 h and then cooled to room temperature. Upon addition of NH_4PF_6 (80 mg), a precipitate was formed and washed with water to give a white product (7.1 mg, 90%). ^1H NMR (500 MHz, CD_3CN , 298 K, δ): 8.75 (s, 24H, $\text{tpy-H}^{3,5}$), 8.60 (d, $J = 8.1$ Hz, 24H, $\text{tpy-H}^{3,3'}$), 8.05 (td, $J = 7.7, 1.6$ Hz, 24H, $\text{tpy-H}^{4,4'}$), 7.70 (d, $J = 4.8$ Hz, 24H, $\text{tpy-H}^{6,6'}$), 7.59 (d, $J = 7.9$ Hz, 24H, Ph- H^p), 7.47 (m, 72H, Ph- H^j , Ph- H^i , and Ph- H^n), 7.41 (dd, $J = 8.4, 1.9$ Hz, 12H, Ph- H^c), 7.31

(dd, $J = 8.3, 1.9$ Hz, 12H, Ph- H^a), 7.28–7.25 (m, 36H, Ph- H^d and Ph- H^b), 7.23 (dd, $J = 7.4, 5.2$ Hz, 24H, tpy- $H^{5,5'}$), 7.16 (m, 36H, Ph- H^g and Ph- H^f). ^{13}C NMR (125 MHz, CD_3CN , 298 K, δ) 155.91, 149.70, 147.94, 147.72, 144.17, 141.10, 140.25, 139.42, 138.54, 138.29, 136.82, 135.71, 134.17, 131.84, 131.83, 130.98, 130.76, 130.42, 128.53, 128.15, 127.65, 127.52, 126.07, 125.03, 124.84, 122.99, 121.31. ESI-MS (m/z): 1597.9 $[\text{M}-5\text{PF}_6]^{5+}$ (calcd: 1597.9), 1307.6 $[\text{M}-6\text{PF}_6]^{6+}$ (calcd: 1307.6), 1099.8 $[\text{M}-7\text{PF}_6]^{7+}$ (calcd: 1099.8), 944.4 $[\text{M}-8\text{PF}_6]^{8+}$ (calcd: 944.4), 823.4 $[\text{M}-9\text{PF}_6]^{9+}$ (calcd: 823.4).

Results and Discussion

Synthesis, characterization, and interlocking study of complex **SA**

The attempt to synthesis ligand **LAH** (without alkyl chain) was unsuccessful because of the poor solubility of **LAH**. To improve the solubility, alkyl chains were introduced into ligand **LA**. The ^1H NMR spectra of ligand **LA** and corresponding complex **SA** exhibited one set of signals (Figures 1a and 1b, Supporting Information Figures S17 and S22), indicating the formation of a highly symmetrical structure.⁷⁹ Compared with ligand **LA**, the 3',5'-tpy protons of **SA** were shifted downfield due to the lower electron density after complexation,⁸⁰ while the

protons at the 6,6'' position of tpy were significantly shifted upfield (~ 0.92 ppm) based on the electron shielding effect.⁶⁴ The full assignments of ^1H NMR were confirmed by detailed two-dimensional (2D) COSY (correlation spectroscopy). All NMR (^1H , ^{13}C , and COSY) and MALDI-TOF data of ligand **LA** and complex **SA** were shown in Supporting Information Figures S17–28.

Interestingly, the ^1H NMR spectra changed significantly upon increasing the concentration of **SA**. Specifically, all protons on tpy gradually shifted upfield with increasing concentration from 0.5 to 5.5 mg/mL (Figures 1b–1g, and Supporting Information Figures S23 and S24), and a set of mixed signals appeared during this process (Figures 1d and 1e). Notably, the protons of the phenyl moieties on the **LA** axis (H^k , H^l , H^m , H^n , and H^o) exhibited a significant upfield shift compared with the other protons. Moreover, the protons on the corner phenyl group displayed an opposite trend: H^b gradually shifted upfield, while H^a moved downfield. Meanwhile, $H^{c'}$ and H^c presented a similar trend compared with H^a and H^b . These phenomena indicated that as the concentration increased, there existed obvious interactions between **SA** that influenced the electron cloud density. Furthermore, the diffusion-ordered NMR spectroscopy (DOSY) experiment was employed to measure the trend of the size change as further evidence; we observed a single band ($\log D = -9.38$) at a lower concentration and another band ($\log D = -9.47$) at a higher concentration. The experimental hydrodynamic radii (r_h) calculated via the Stokes–Einstein equation were 1.4 nm at 0.5 mg/mL and 1.8 nm at 5.5 mg/mL (Supporting Information Figures S29 and S30 and Table S2),⁸¹ suggesting an interlocked structure formed by two or more **SA** (Scheme 1) at high concentration. In addition, considering the steric hindrance of tpy, it is reasonable to speculate that the phenyl rings on the axis were stacked through π – π interaction to form interlocked structures, instead of the ring-in-ring assemblies.

ESI-MS was used to detect the detailed molecular compositions of **SA**. In Figure 2a, only one prominent set of peaks with different charge states from 2+ to 4+ was observed at a concentration of 0.5 mg/mL due to the successive loss of a different number of PF_6^- . The isotope pattern of each peak matched well with the simulated isotope pattern of **SA** (Figure 2a and Supporting Information Figure S1) for the desired dimer with a molecular weight of 3805.3 Da. Upon concentrating **SA** (>0.5 mg/mL), a new peak at m/z 1379.1 for 5+ was observed (Figure 2b), which was exactly twice the molecular weight of **SA**, confirming the formation of [2]catenane (**SA-2**). At a higher concentration, the appearance of the peak at m/z of 2141.2 for 5+ further confirmed the formation of [3]catenane (**SA-3**) (Figure 2c and Supporting Information Figure S2).

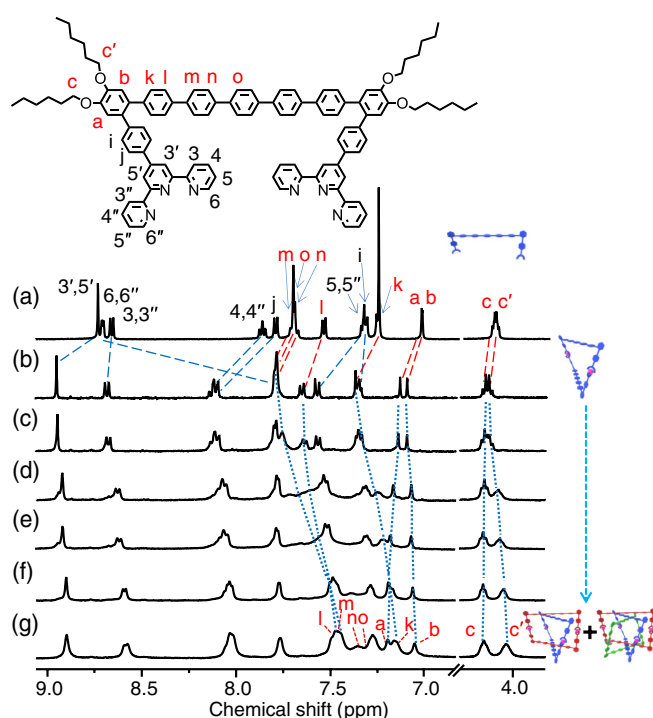


Figure 1 ^1H NMR (a–g) spectra (500 MHz, 298 K) of (a) **LA** in CDCl_3 , (b) **SA**-0.5 mg/mL, (c) **SA**-1.5 mg/mL, (d) **SA**-2.5 mg/mL, (e) **SA**-3.5 mg/mL, (f) **SA**-4.5 mg/mL, and (g) **SA**-5.5 mg/mL in CD_3CN .

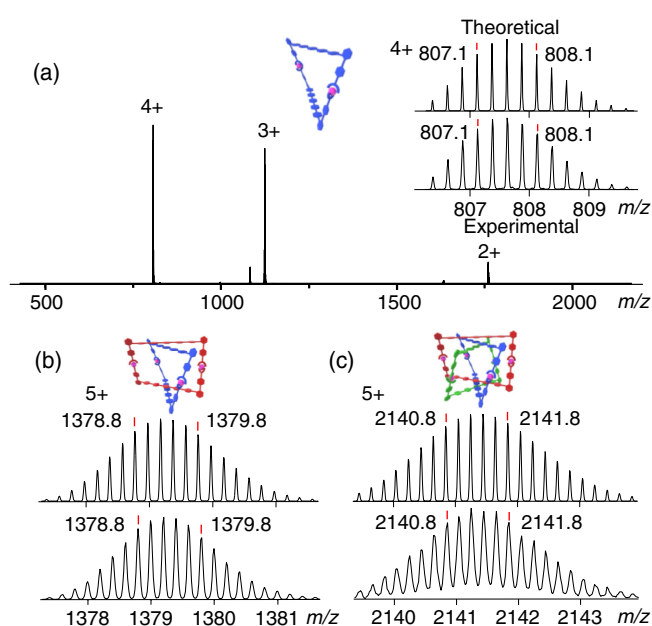


Figure 2 | (a) ESI-MS spectrum of complex **SA** at a concentration of 0.5 mg/mL. Calculated (top) and measured (bottom) isotope patterns for (b) [2]catenane (**SA-2**) and (c) [3]catenane (**SA-3**) at high concentration (PF_6^- as counterion).

Synthesis, characterization, and interlocking study of complexes **SB** and **SBH**

To further explore the effects of space constraints on the structure and function of the supramolecules, complex **SB**, based on tritopic ligand **LB**, was obtained through the same synthetic method as shown in Supporting Information Scheme S2. The ^1H NMR spectrum of **SB** also showed one set of signals with similar chemical shifts to **SA** (Supporting Information Figure S36). The protons at tpy-6,6'' shifted upfield (~ 1.06 ppm) characteristically upon complexation because of the electron shielding effect, while almost all other peaks were shifted downfield. The assignments of other proton peaks were confirmed through the 2D COSY experiment. All NMR (^1H , ^{13}C , and COSY) and MALDI-TOF data of ligand **LB** and corresponding complex **SB** were shown in Supporting Information Figures S31–S41.

The ESI-MS spectrum of complex **SB** displayed a series of continuous charge states from 3+ to 6+ (Supporting Information Figure S3). The experimental m/z values and isotope patterns agreed well with the simulated results of **SB** with a molecular weight of 5637.2 Da. To detect the possible presence of overlapping isomers or conformers, TWIM-MS was introduced as an advanced level of MS analysis. As shown in Supporting Information Figure S14, each charge state showed a narrow drift time distribution, indicating no conformers or overlapping isomers of the assembly existed in this complex. We further

investigated the concentration dependence of **SB** by using ^1H NMR spectroscopy. The experimental results showed one set of signals at a concentration of <10 mg/mL. Interestingly, when the concentration exceeded 10 mg/mL, some peaks attributed to interlocked structures (named **SB-2**) appeared after 3 days (Supporting Information Figures S6, S13, and S37) because the alkyl chains in complex **SB** acted as a sterically hindered group to inhibit the formation of interlocked structures.⁸² To further confirm this speculation, a new tritopic ligand **LBH** (without the alkyl chain) was synthesized and assembled in the same process to obtain complex **SBH**. At a relatively low concentration (≤ 3 mg/mL), the ^1H NMR spectrum of **SBH** (Figures 3a and 3b) also showed one set of peaks. The ligand **LBH** and corresponding complex **SBH** were fully characterized by NMR (^1H , ^{13}C , COSY, and DOSY) and MALDI-TOF (Supporting Information Figures S42–S56). In addition, the ESI-MS spectrum of **SBH** displayed a set of peaks ranging from 3+ to 6+ with the molecular weight of 4435.7 Da (Supporting Information Figure S4 and S11). However, the other two sets of peaks were observed at a higher concentration (≥ 3 mg/mL) (Figure 3c). Moreover, with increasing the concentration, the appearance of

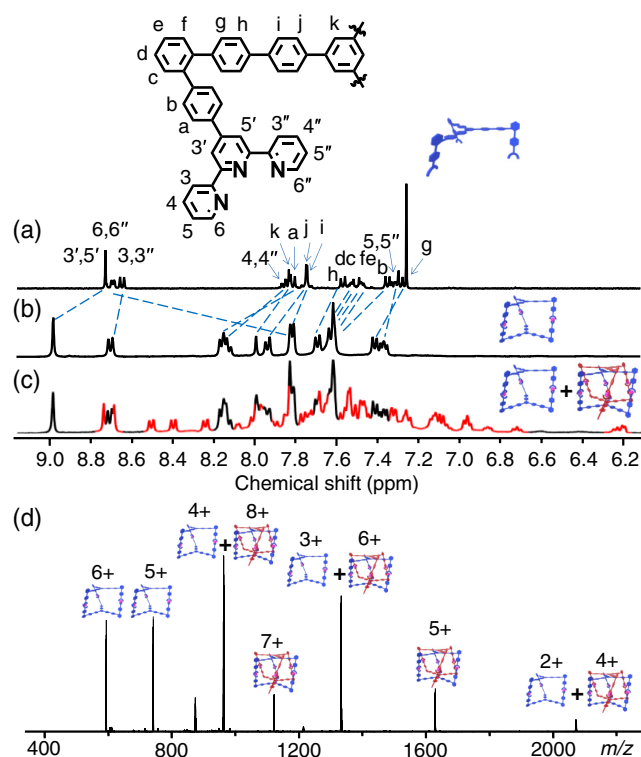


Figure 3 | ^1H NMR (a–c) spectra (500 MHz, 298 K) of (a) **LBH** in CDCl_3 , (b) complex **SBH** (0.5 mg/mL), and (c) complexes **SBH** and [2]catenane (**SBH-2**) (5 mg/mL) in CD_3CN . (d) ESI-MS spectrum of complexes **SBH** and [2]catenane (**SBH-2**) at 5 mg/mL.

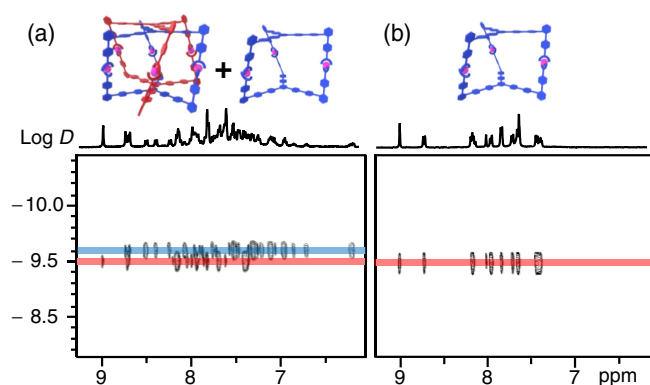


Figure 4 | (a) 2D DOSY NMR spectra (298 K, CD_3CN) of the mixed complexes **SBH** and interlocked **SBH-2** (5 mg/mL), (b) complex **SBH** (0.5 mg/mL).

new peaks at 2072.5 (4+), 1629.0 (5+), 1333.2 (6+), 1121.9 (7+), and 964.0 (+8) demonstrated the formation of another structure (Figure 3d, Supporting Information Figures S5 and S12). After deconvolution, the molecular weight of the assembly was 8871.4 Da, which was exactly twice the molecular weight of **SBH**, suggesting the formation of the interlocked structure **SBH-2**.

The 2D DOSY NMR spectrum of **SBH** are shown in Figure 4b. One single band ($\log D = -9.49$) with a diffusion coefficient (D) around $3.21 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ at low concentration was observed, while another band ($\log D = -9.58$) with a diffusion coefficient around $2.76 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ emerged as the concentration increased (Figure 4a). The hydrodynamic radius was calculated by the Stokes-Einstein equation. Compared with **SBH**, the hydrodynamic radius of interlocked complex **SBH-2** (2.2 nm) was only 0.3 nm larger than that of **SBH** (1.9 nm), which indicated that the interpenetrated cage may be formed by the π - π stacking between two **SBH** upper surface at higher concentration (as shown by Scheme 1).

Synthesis, characterization, and host-guest study of complexes **SC** and **SCH**

Given the successful assembly of complexes **SA** and **SB**, hexatopic tpy ligands **LC** and **LCH** (Figure 5a, Supporting Information Figure S52, and Scheme S2) were synthesized, and the corresponding complexes (**SC** and **SCH**) were obtained by the same self-assembly method. NMR (^1H , ^{13}C , COSY, and DOSY) and MALDI-TOF of ligands (**LC** and **LCH**) and corresponding complexes (**SC** and **SCH**) were shown in Supporting Information Figures S52–S70. The ^1H NMR spectra of these two complexes exhibited one set of signals (Figure 5b and Supporting Information Figure S57), which were all assigned by 2D COSY (Supporting Information Figures S60 and S69). Furthermore, ESI-MS (Figure 5c and Supporting Information Figures S7, S8, S15, and S16) and TWIM-MS

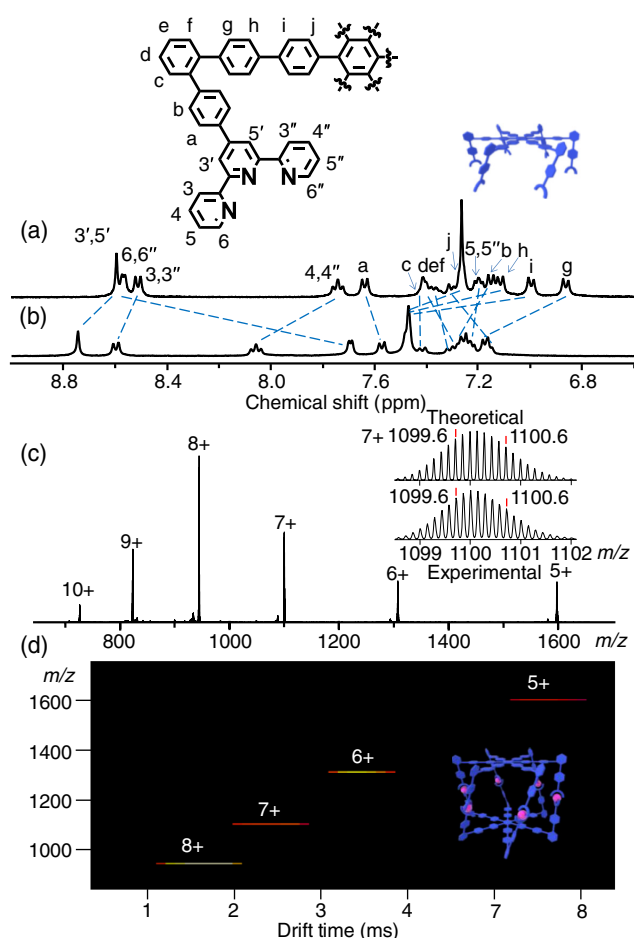


Figure 5 | ^1H NMR spectra (500 MHz, 298 K) of (a) **LCH** in CDCl_3 and (b) complex **SCH** in CD_3CN . (c) ESI-MS spectra and (d) 2D TWIM-MS plot of complex **SCH**.

(Figure 5d and Supporting Information Figure S15) were employed to analyze the compositions of these two complexes. The results proved the formations of discrete hexagonal prisms with a molecular weight of 11,120.3 Da for **SC** and 8714.3 Da for **SCH**. 2D DOSY NMR of these two complexes in CD_3CN (Supporting Information Figures S61 and S70) clearly showed a single band ($\log D = -9.43$), which demonstrated a similar scale of **SC**, **SA**, and **SB** at the single-molecular condition (Supporting Information Table S2). For **SC** and **SCH**, even if we increased the concentration and extended the time, interlocked structures were not observed in either NMR or ESI-MS spectra (Supporting Information Figures S59 and S67), which is probably due to the significantly enhanced space constraints of the twisted hexagonal prisms.

To verify this speculation, the single crystal of **SCH** was successfully obtained through slowly diffusing the vapor of diethyl ether into **SCH** in acetonitrile at 15 °C for 4 weeks. As shown in Figure 6, the single-crystal structure of **SCH** consisted of two parallel hexatopic ligands **LCH** and six edges with Zn(II) ions and showed a

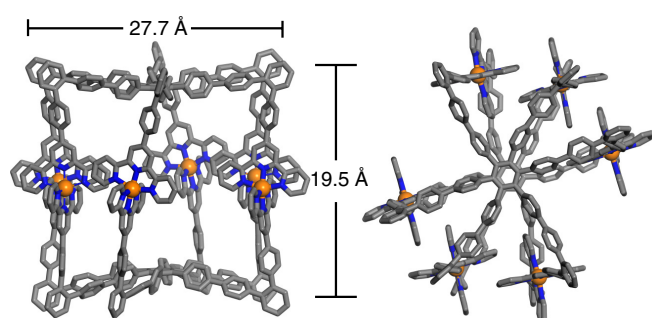


Figure 6 | X-ray crystal structure of **SCH**. Color code: C, grey; N, blue; Zn, orange. Front (left) and top view of **SCH** (right). Hydrogen atoms and counterions were omitted for the sake of clarity.

relatively congested structure, which confirmed our speculation about space constraints influencing molecular interlocking. The junction angle between the panel and tpy was 63° , indicating the twisting of aromatic panels. The diameter and height of the panel were 27.72 and 19.52 Å, respectively. The detailed crystal parameters were shown in Supporting Information Table S1. This indicated that the **SA**, **SB**, and **SBH** (with similar structures as **SCH**) could endow enough space to form the interlocked structures such as [2]catenane and [3]catenane. The molecular modeling of corresponding complexes in Supporting Information Figures S71–S73.

Due to the significant effect of space constraints, **SCH** could not form an interlocked structure like **SA** and **SBH**. However, the single-crystal structure of **SCH** clearly showed a certain cavity in the cage and large π -conjugated aromatic panels. Notably, the steric hindrance of tpy could provide essential help to enclose the cavity, which enabled the exploration of the encapsulations for the guest molecules. In the study of the encapsulation for guest molecules in supramolecular cavities, the previous works have mainly focused on the regular structures, such as polyaromatic hydrocarbons and fullerenes.⁵¹ However, the molecular encapsulations for more complex conformation and diverse architectures, such as calixarene, pillararene, and others, were challenging and rarely reported.^{83,84} Taking into account the special shape of the cavity and the significant space constraints, **SCH** may have the potential to encapsulate calixarene molecules selectively.

To promote the dissolution of calix[*n*]arene (*n* = 4, 6, and 8) at room temperature, a mixture of CD_3CN and CDCl_3 (3:1) was used as the solvent. We found that by adding excessive calix[4]arene and calix[6]arene in **SCH**, respectively, the hydrogen signals of **SCH** (Figures 7a) exhibited a certain shift (Figures 7b, 7c, and Supporting Information Figure S77), especially in the latter case, suggesting successful encapsulations. But the addition of calix[8]arene did not change the hydrogen

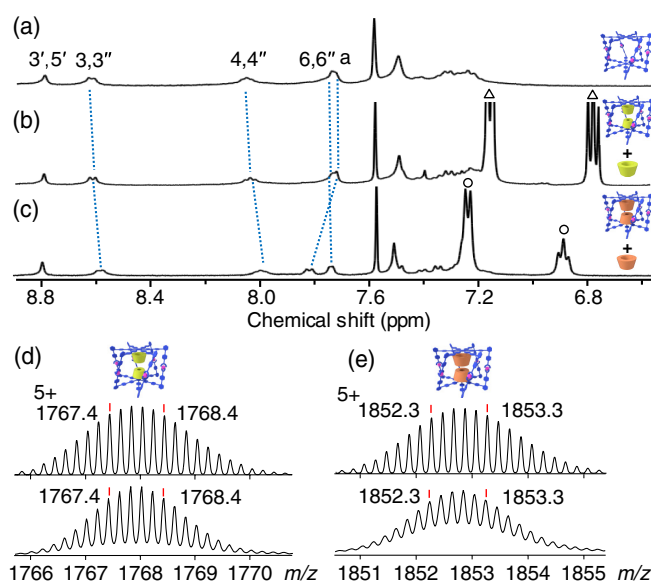


Figure 7 | ^1H NMR (a–c) spectra (500 MHz, $\text{CD}_3\text{CN}:\text{CDCl}_3$ 3:1, 298 K) of (a) **SCH**, (b) calix[4]arene $\subset$ **SCH**, and (c) calix[6]arene $\subset$ **SCH**; Measured (bottom) and calculated (top) isotope patterns for (d) $[2 \text{ calix[4]arene} \subset \text{SCH-5PF}_6^-]^{5+}$ and (e) $[2 \text{ calix[6]arene} \subset \text{SCH-5PF}_6^-]^{5+}$ (Δ , free guest peaks of calix[4]arene; $\circ$, free guest peaks of calix[6]arene).

environment of **SCH** (Supporting Information Figure S74b), probably due to the relatively large size and poor solubility of calix[8]arene. Furthermore, the ESI-MS spectrum showed a set of peaks ranging from 4+ to 8+ with the molecular weight of 9564.3 Da (Figure 7d and Supporting Information Figure S76) and another set of peaks from 5+ to 8+ with the molecular weight of

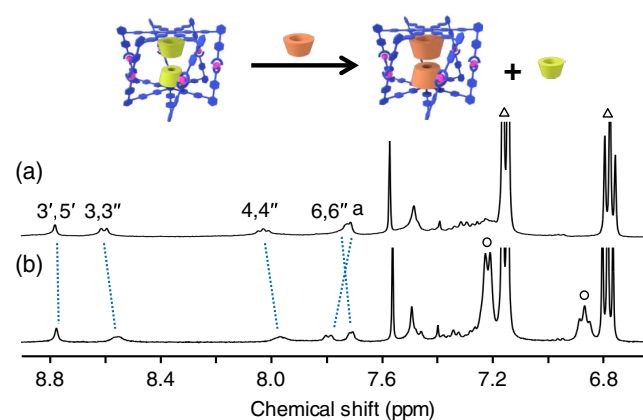


Figure 8 | ^1H NMR spectra (500 MHz, $\text{CD}_3\text{CN}:\text{CDCl}_3$ 3:1, 298 K) of (a) calix[4]arene $\subset$ **SCH** and (b) competitive encapsulation of calix[6]arene in calix[4]arene $\subset$ **SCH** (Δ , free guest peaks of calix[4]arene; $\circ$, free guest peaks of calix[6]arene).

9988.8 Da (Figure 7e and Supporting Information Figure S79), indicating two calix[4]arene or two calix[6]arene were encapsulated in **SCH**. Interestingly, when these two types of guest molecules were together added to **SCH** (Supporting Information Figure S80c) or calix[6]arene (Figure 8b) was added after calix[4]arene encapsulation (Figure 8a), the ¹H NMR and ESI-MS spectra only showed the signal of the encapsulated calix[6]arene, indicating that only calix[6]arene were selectively encapsulated into **SCH** (Supporting Information Figure S81b). These results illustrated that **SCH** had a better affinity for calix[6]arene than calix[4]arene. (Supporting Information Figures S74–S81).

Conclusion

Three types of tpy ligands with identical diameters and different space constraints were successfully synthesized and further assembled into the twisted dimeric architectures. Interlocked structures [2]catenane (**SA-2**) and [3]catenane (**SA-3**) were observed by increasing the concentration of **SA**. Moreover, interlocked cages **SB-2** and **SBH-2** were also observed with increases in the concentrations of **SB** and **SBH**. This interlocked feature was affected by the steric hindrance of the alkyl chains. For complexes **SC** and **SCH**, constructed by hexatopic ligands **LC** and **LCH** with the maximum space constraints, the interlocking phenomena were not observed. However, the enhanced space restriction effect endowed **SCH** a well-sealed cavity for the recognition of guest molecules. In particular, it could selectively encapsulate calix[6]arene from the mixture of calix[4]arene and calix[6]arene. Our design realized the transformation from interlocked structures to host-guest systems by adjusting space constraints and provided a novel strategy for the development of interlocked structures and host-guest chemistry.

Supporting Information

Supporting Information is available and includes the synthesis of the ligands and complexes, NMR spectra, MALDI-TOF spectra, ESI-MS data, and single-crystal data.

Conflict of Interest

The authors declare no conflict of interest.

Acknowledgments

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